**(i)** Calculate the e.m.f. $E$ of the battery.

$E$ = ...................................................... V [2]

**(ii)** The filament wire of lamp B has a cross-sectional area of $1.4 \times 10^{-9}\,m^2$. The number of free (conduction) electrons per unit volume in the metal of the filament wire is $3.4 \times 10^{28}\,m^{-3}$.

Calculate the average drift speed of the free electrons in the filament wire of lamp B.

average drift speed = ................................................ $m\,s^{-1}$ [3]

[Total: 8]